Experiment 1: Install Matplotlib and Display a Simple Line Graph

Python Program

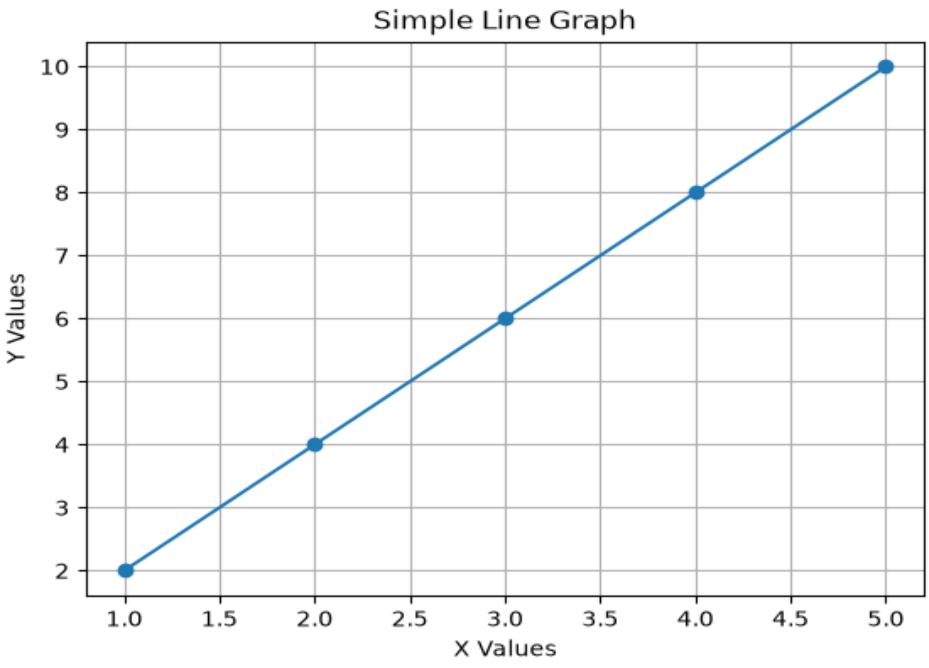
```
import matplotlib.pyplot as plt


x = [1, 2, 3, 4, 5]
y = [2, 4, 6, 8, 10]


plt.plot(x, y, marker="o")
plt.title("Simple Line Graph")
plt.xlabel("X Values")
plt.ylabel("Y Values")
plt.grid(True)

plt.show()
```

Output



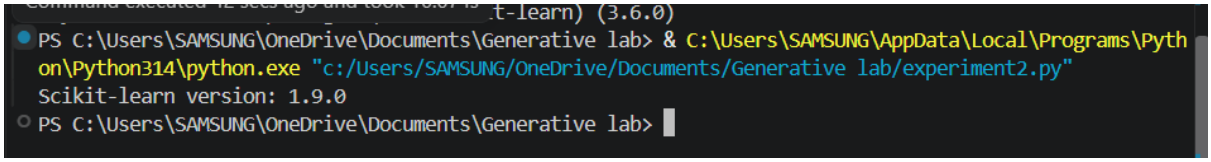
Experiment 2: Install Scikit-learn and Print Its Version

Python Program

```
import sklearn

print("Scikit-learn version:", sklearn.__version__)
```

Output



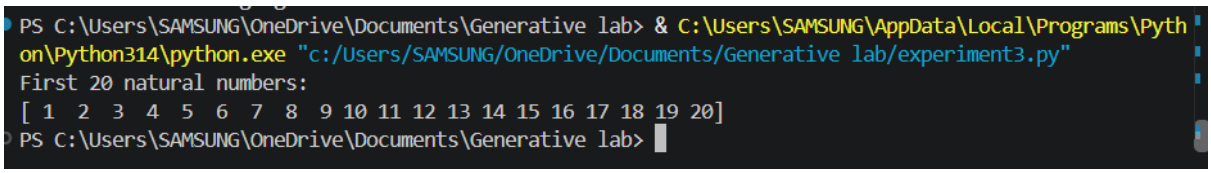
Experiment 3: Create a NumPy Array Containing the First 20 Natural Numbers

Python Program

```
import numpy as np


arr = np.arange(1, 21)
print("First 20 natural numbers:")
print(arr)
```

Output



Experiment 4: Create a Pandas DataFrame with Student Names and Grades

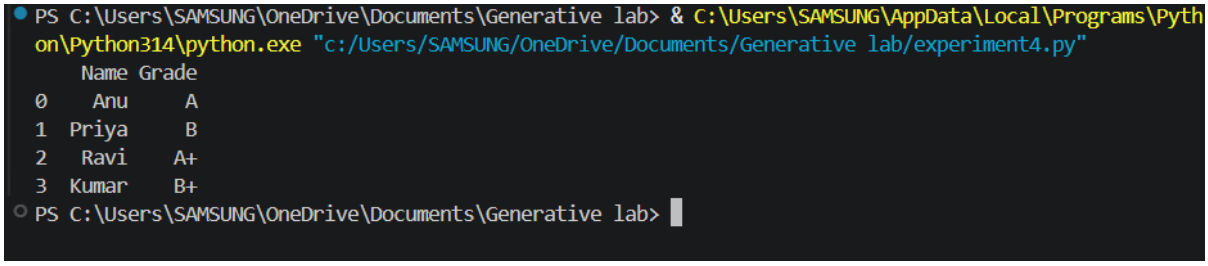
Python Program

```
import pandas as pd

data = {
    "Name": ["Anu", "Priya", "Ravi", "Kumar"],
    "Grade": ["A", "B", "A+", "B+"]
}

df = pd.DataFrame(data)
print(df)
```

Output



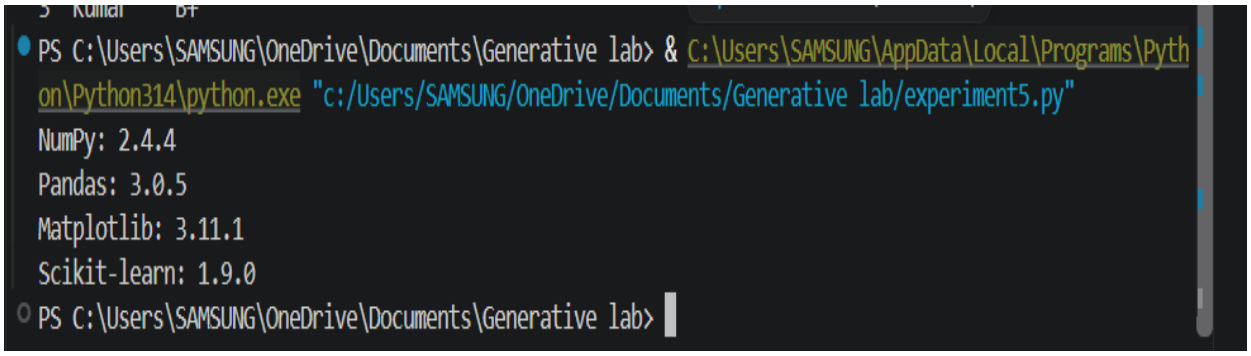
Experiment 5: Verify the Installed Versions of All AI Libraries on Your System

Python Program

```
import numpy as np
import pandas as pd
import matplotlib
import sklearn

print("NumPy:", np.__version__)
print("Pandas:", pd.__version__)
print("Matplotlib:", matplotlib.__version__)
print("Scikit-learn:", sklearn.__version__)
```

Output



Experiment 6: Create Tensors Using PyTorch and Perform Basic Mathematical Operations

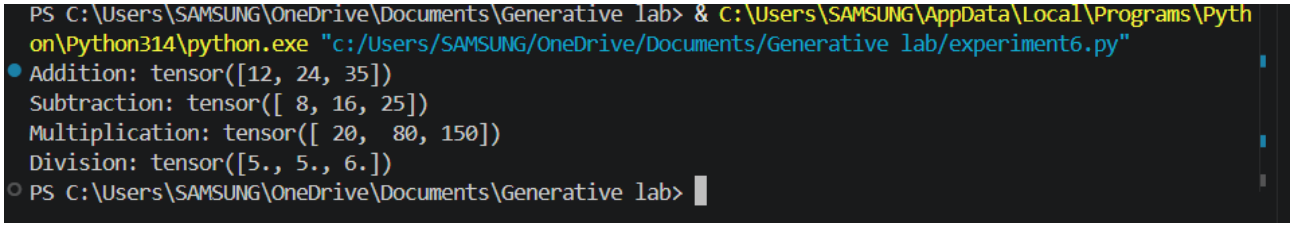
Python Program

```
import torch

a = torch.tensor([10, 20, 30])
b = torch.tensor([2, 4, 5])

print("Addition:", a + b)
print("Subtraction:", a - b)
print("Multiplication:", a * b)
print("Division:", a / b)
```

Output



Experiment 7: Load a Pre-trained Transformer Model Using Hugging Face and Perform Sentiment Analysis

Python Program

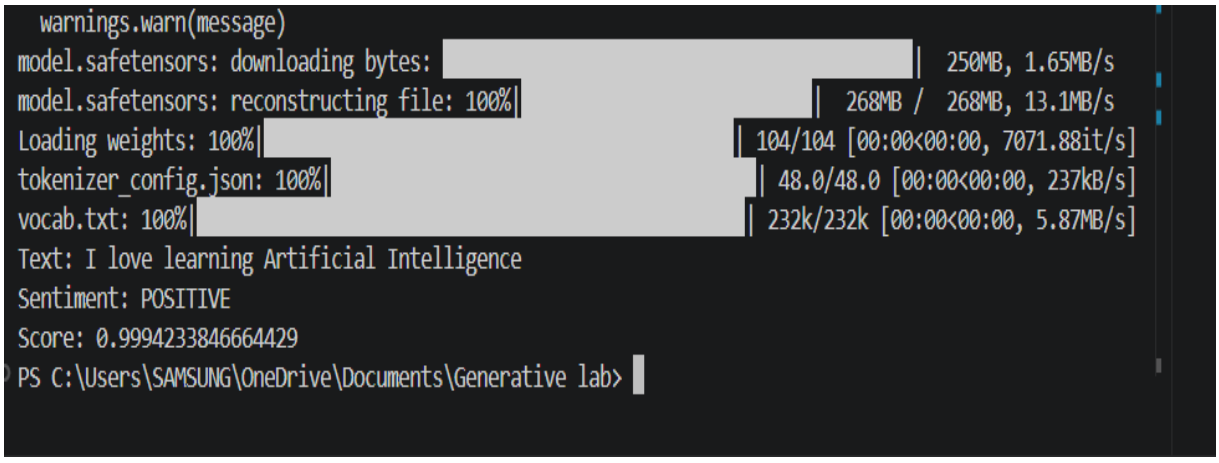
```
from transformers import pipeline

sentiment = pipeline("sentiment-analysis")

text = "I love learning Artificial Intelligence"
result = sentiment(text)

print("Text:", text)
print("Sentiment:", result[0]["label"])
print("Score:", result[0]["score"])
```

Output



Experiment 8: Tokenization Using BERT Tokenizer

Python Program

```
from transformers import BertTokenizer

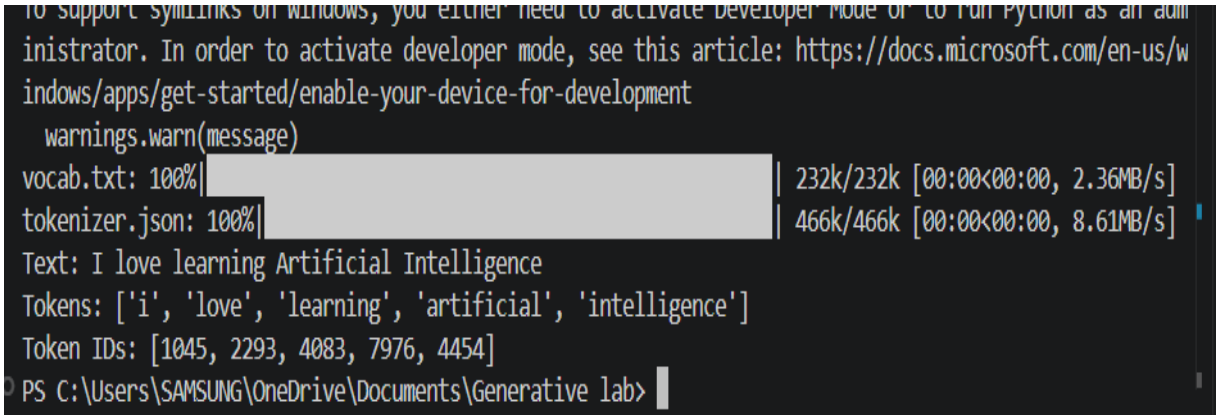
tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")

text = "I love learning Artificial Intelligence"

tokens = tokenizer.tokenize(text)
ids = tokenizer.convert_tokens_to_ids(tokens)

print("Text:", text)
print("Tokens:", tokens)
print("Token IDs:", ids)
```

Output



Experiment 9: Generate Text Using a Pre-trained GPT-2 Model

Python Program

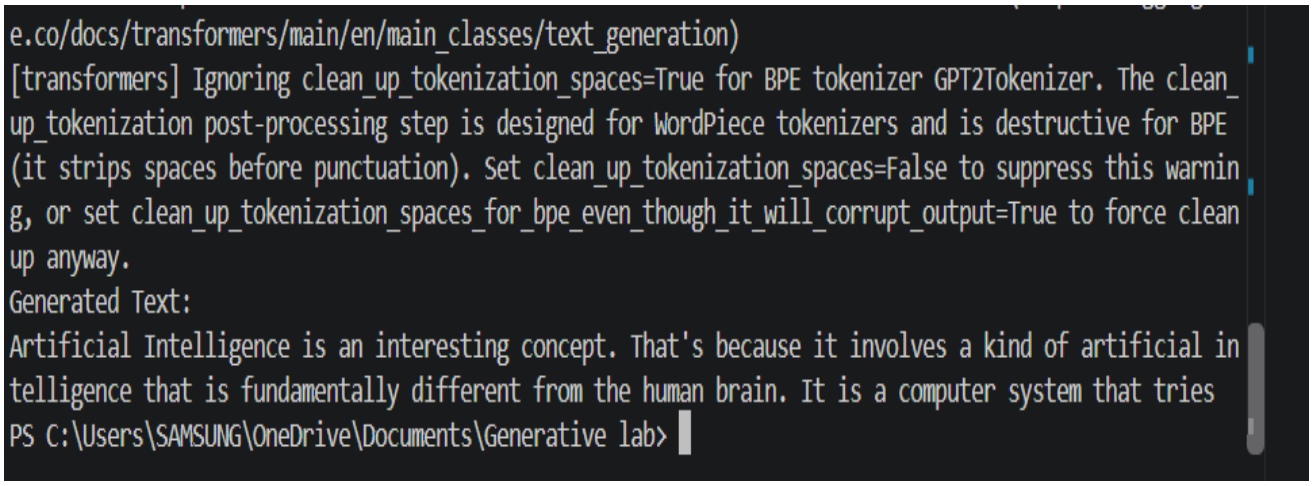
```
from transformers import pipeline

generator = pipeline("text-generation", model="gpt2")

prompt = "Artificial Intelligence is"
result = generator(prompt, max_new_tokens=30, num_return_sequences=1)

print("Generated Text:")
print(result[0]["generated_text"])
```

Output



Experiment 10: Sentiment Analysis Using Hugging Face Pipeline API

Python Program

```
from transformers import pipeline

model = pipeline("sentiment-analysis")

text = "This movie is very good"

result = model(text)

print("Text:", text)

print("Sentiment:", result[0]["label"])

print("Score:", result[0]["score"])
```

Output

```
on\Python314\python.exe "c:/Users/SAMSUNG/OneDrive/Documents/Generative lab/experiment10.py"
[transformers] No model was supplied, defaulted to distilbert/distilbert-base-uncased-finetuned-sst-2
-english and revision 714eb0f.
Using a pipeline without specifying a model name and revision in production is not recommended.
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable high
er rate limits and faster downloads.
Loading weights: 100%|                                                  | 104/104 [00:00<00:00, 12238.93it/s]
Text: This movie is very good
Sentiment: POSITIVE
Score: 0.9998663663864136
PS C:\Users\SAMSUNG\OneDrive\Documents\Generative lab>
```

Experiment 11: Load the BERT Tokenizer and Tokenize a Sentence

Python Program

```
from transformers import BertTokenizer

tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")

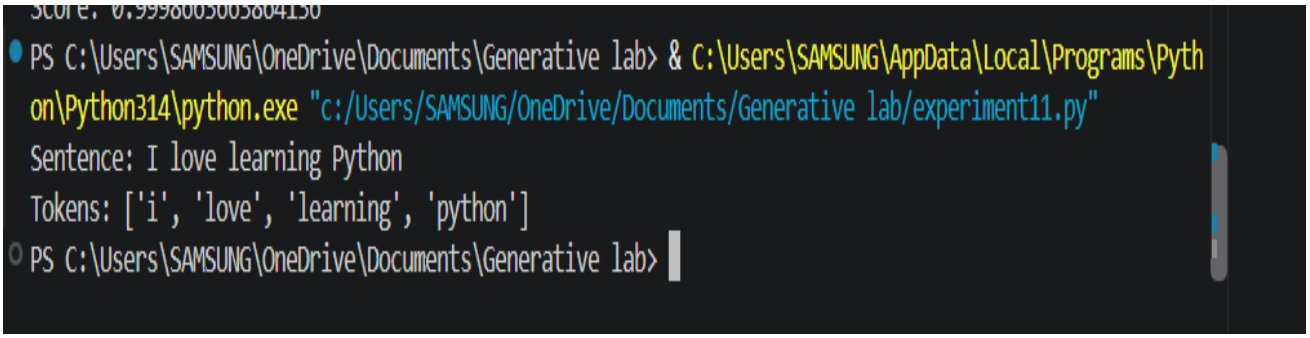
text = "I love learning Python"

tokens = tokenizer.tokenize(text)

print("Sentence:", text)

print("Tokens:", tokens)
```

Output



Experiment 12: Load a Pre-trained BERT Model and Generate Contextual Embeddings

Python Program

```
import torch
from transformers import BertTokenizer, BertModel

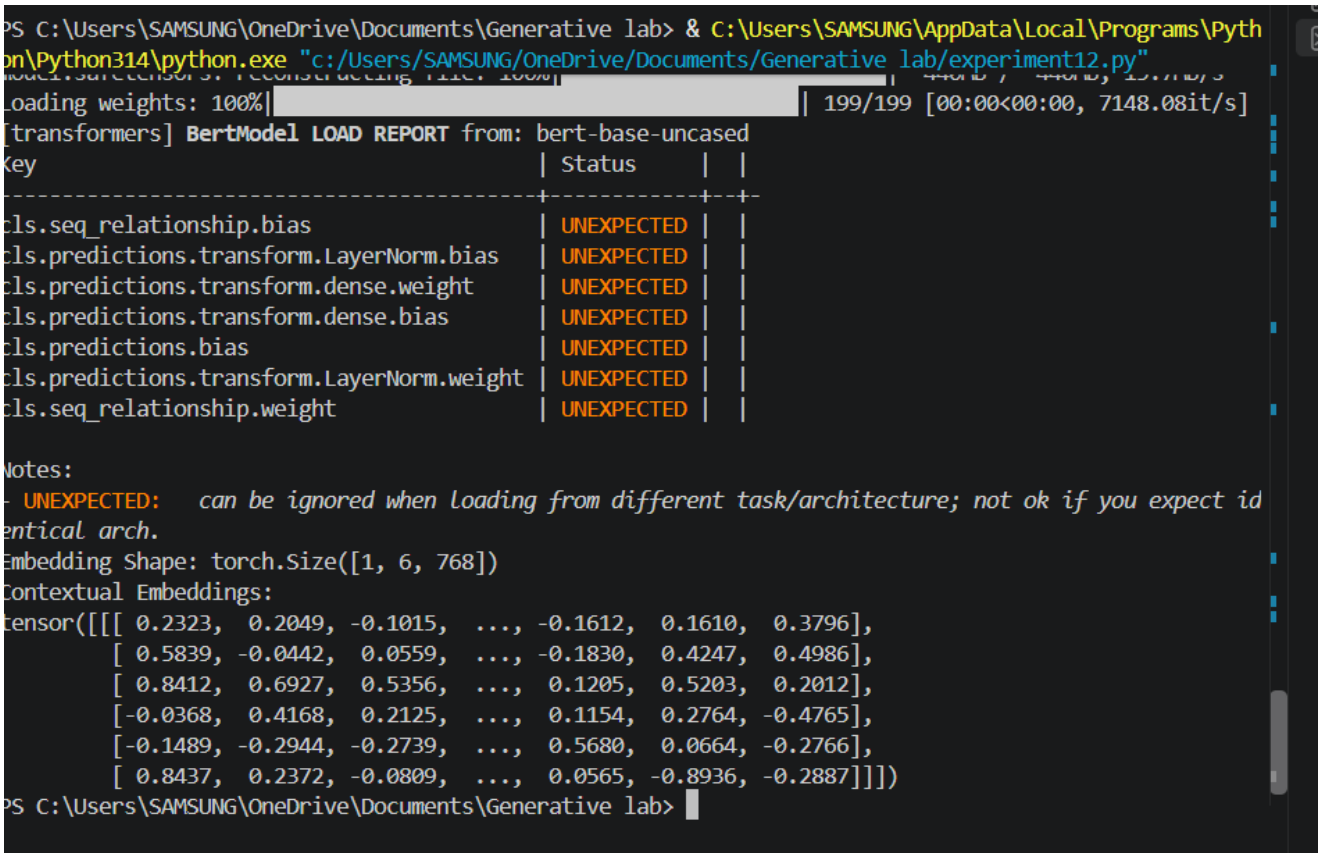
tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")
model = BertModel.from_pretrained("bert-base-uncased")

text = "I love learning Python"
inputs = tokenizer(text, return_tensors="pt")

with torch.no_grad():
    outputs = model(**inputs)

print("Embedding Shape:", outputs.last_hidden_state.shape)
print("Contextual Embeddings:")
print(outputs.last_hidden_state)
```

Output


PS C:\Users\SAMSUNG\OneDrive\Documents\Generative lab> & C:\Users\SAMSUNG\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/SAMSUNG/OneDrive/Documents/Generative lab/experiment12.py"
Loading weights: 100%| | 199/199 [00:00<00:00, 7148.08it/s]
[transformers] BertModel LOAD REPORT from: bert-base-uncased
Key | Status | |
cls.seq_relationship.bias | UNEXPECTED | |
cls.predictions.transform.LayerNorm.bias | UNEXPECTED | |
cls.predictions.transform.dense.weight | UNEXPECTED | |
cls.predictions.transform.dense.bias | UNEXPECTED | |
cls.predictions.bias | UNEXPECTED | |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED | |
cls.seq_relationship.weight | UNEXPECTED | |
Notes:
- UNEXPECTED: can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Embedding Shape: torch.Size([1, 6, 768])
Contextual Embeddings:
tensor([[[0.2323, 0.2049, -0.1015, ..., -0.1612, 0.1610, 0.3796],
[0.5839, -0.0442, 0.0559, ..., -0.1830, 0.4247, 0.4986],
[0.8412, 0.6927, 0.5356, ..., 0.1205, 0.5203, 0.2012],
[-0.0368, 0.4168, 0.2125, ..., 0.1154, 0.2764, -0.4765],
[-0.1489, -0.2944, -0.2739, ..., 0.5680, 0.0664, -0.2766],
[0.8437, 0.2372, -0.0809, ..., 0.0565, -0.8936, -0.2887]]])
PS C:\Users\SAMSUNG\OneDrive\Documents\Generative lab>

Experiment 13: Generate Text Using a Pre-trained GPT-2 Model

Python Program

```
from transformers import pipeline

generator = pipeline("text-generation", model="gpt2")

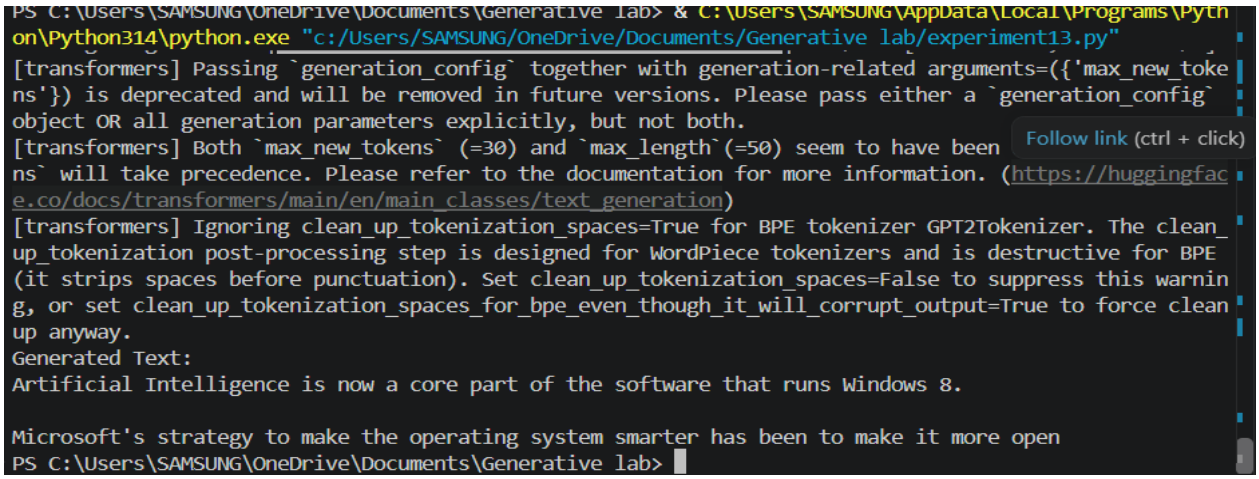
text = "Artificial Intelligence is"

result = generator(text, max_new_tokens=30)

print("Generated Text:")

print(result[0]["generated_text"])
```

Output



Experiment 14: Question Answering Using a Pre-trained Transformer

Python Program

```
from transformers import pipeline

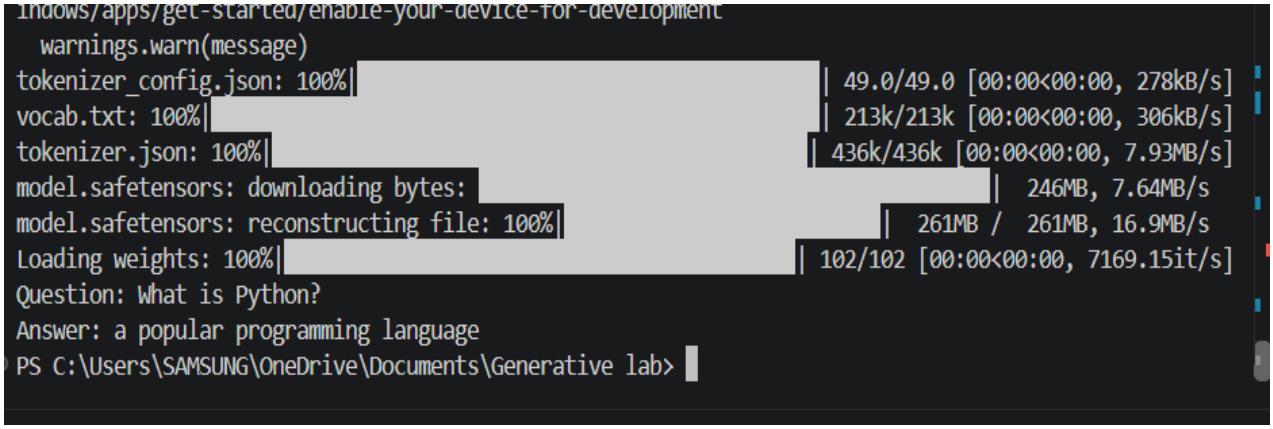
qa = pipeline("question-answering")

context = "Python is a popular programming language."
question = "What is Python?"

result = qa(question=question, context=context)

print("Question:", question)
print("Answer:", result["answer"])
```

Output

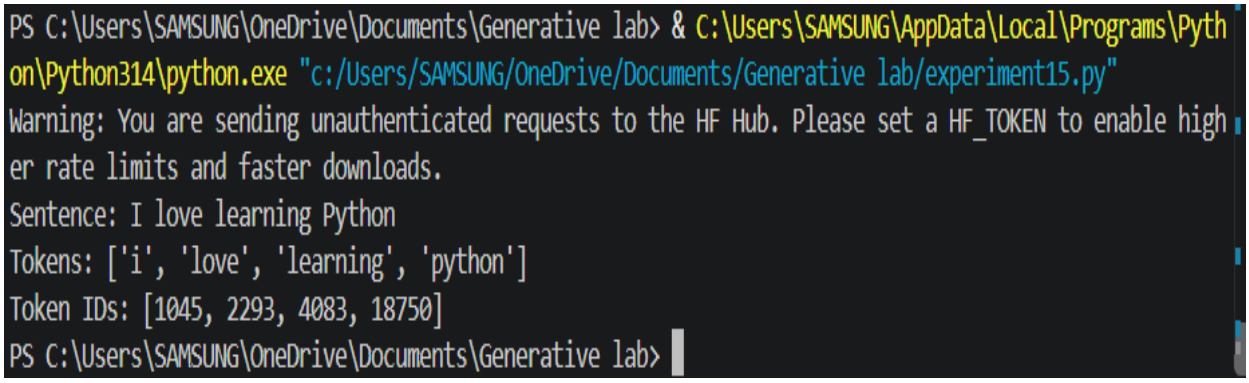


Experiment 15: Tokenize a Sentence Using the BERT Tokenizer and Display Tokens and Token IDs

Python Program

```
from transformers import BertTokenizer
tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")
text = "I love learning Python"
tokens = tokenizer.tokenize(text)
ids = tokenizer.convert_tokens_to_ids(tokens)
print("Sentence:", text)
print("Tokens:", tokens)
print("Token IDs:", ids)
```

Output



Experiment 16: Compare the Tokenization Outputs of BERT and GPT-2

Python Program

```
from transformers import BertTokenizer, GPT2Tokenizer
bert = BertTokenizer.from_pretrained("bert-base-uncased")
gpt2 = GPT2Tokenizer.from_pretrained("gpt2")
text = "I love learning Python"
print("BERT Tokens:", bert.tokenize(text))
print("BERT IDs:", bert.encode(text, add_special_tokens=False))
print("GPT-2 Tokens:", gpt2.tokenize(text))
print("GPT-2 IDs:", gpt2.encode(text))
```

Output

```
PS C:\Users\SAMSUNG\OneDrive\Documents\Generative lab> & C:\Users\SAMSUNG\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/SAMSUNG/OneDrive/Documents/Generative lab/experiment16.py"
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
BERT Tokens: ['i', 'love', 'learning', 'python']
BERT IDs: [1045, 2293, 4083, 18750]
GPT-2 Tokens: ['I', 'Ġlove', 'Ġlearning', 'ĠPython']
GPT-2 IDs: [40, 1842, 4673, 11361]
PS C:\Users\SAMSUNG\OneDrive\Documents\Generative lab>
```

Experiment 17: Generate Contextual Embeddings for a Sentence Using BERT

Python Program

```
import torch
from transformers import BertTokenizer, BertModel

tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")
model = BertModel.from_pretrained("bert-base-uncased")

text = "I love learning Python"
inputs = tokenizer(text, return_tensors="pt")

with torch.no_grad():
    outputs = model(**inputs)

embeddings = outputs.last_hidden_state

print("Sentence:", text)
print("Embedding Shape:", embeddings.shape)
print("Contextual Embeddings:", embeddings)
```

Output

```
PS C:\Users\SAMSUNG\OneDrive\Documents\Generative lab> & C:\Users\SAMSUNG\AppData\Local\Programs\Pyth
on\Python314\python.exe "c:/Users/SAMSUNG/OneDrive/Documents/Generative lab/experiment17.py"

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architecture; not ok if you expect id
entical arch.
Sentence: I love learning Python
Embedding Shape: torch.Size([1, 6, 768])
Contextual Embeddings: tensor([[[ 0.2323,  0.2049, -0.1015,  ..., -0.1612,  0.1610,  0.3796],
         [ 0.5839, -0.0442,  0.0559,  ..., -0.1830,  0.4247,  0.4986],
         [ 0.8412,  0.6927,  0.5356,  ...,  0.1205,  0.5203,  0.2012],
         [-0.0368,  0.4168,  0.2125,  ...,  0.1154,  0.2764, -0.4765],
         [-0.1489, -0.2944, -0.2739,  ...,  0.5680,  0.0664, -0.2766],
         [ 0.8437,  0.2372, -0.0809,  ...,  0.0565, -0.8936, -0.2887]]])
PS C:\Users\SAMSUNG\OneDrive\Documents\Generative lab>
```

Experiment 18: Compare Contextual Embeddings for Two Semantically Similar Sentences

Python Program

```
import torch

from transformers import BertTokenizer, BertModel

from torch.nn.functional import cosine_similarity


tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")

model = BertModel.from_pretrained("bert-base-uncased")


s1 = "I love learning Python"

s2 = "I enjoy studying Python"


def get_embedding(sentence):

    inputs = tokenizer(sentence, return_tensors="pt")

    with torch.no_grad():

        output = model(**inputs)

    return output.last_hidden_state[:, 0, :]


e1 = get_embedding(s1)

e2 = get_embedding(s2)

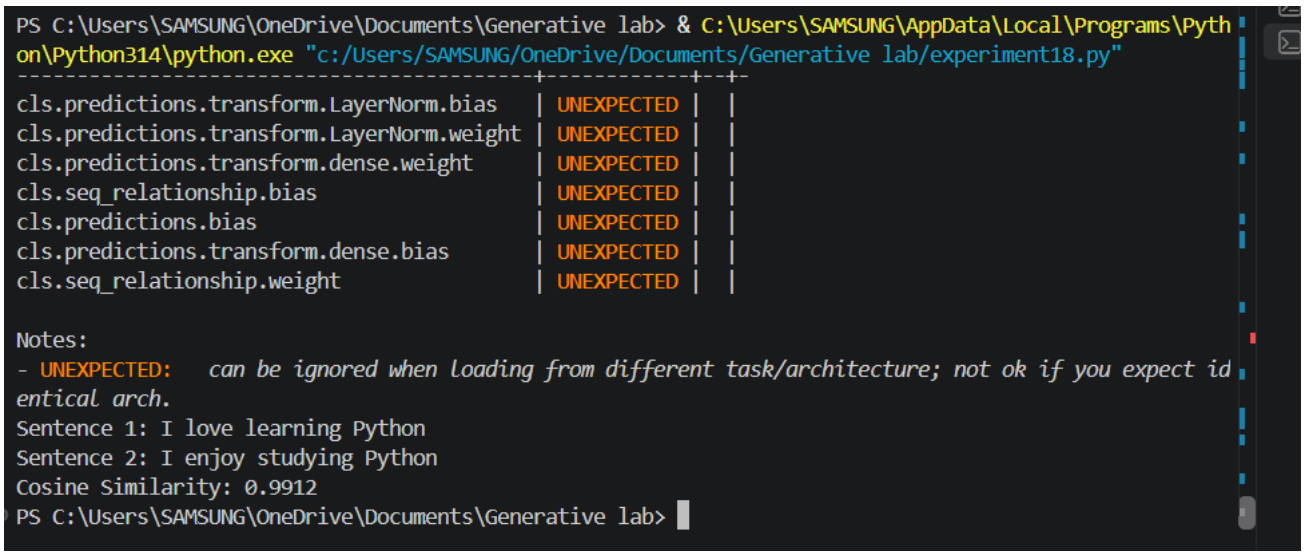

score = cosine_similarity(e1, e2).item()


print("Sentence 1:", s1)
```

```
print("Sentence 2:", s2)

print("Cosine Similarity:", round(score, 4))
```

Output



Experiment 19: Generate Text from a Prompt Using GPT-2

Python Program

```
from transformers import pipeline

generator = pipeline("text-generation", model="gpt2")

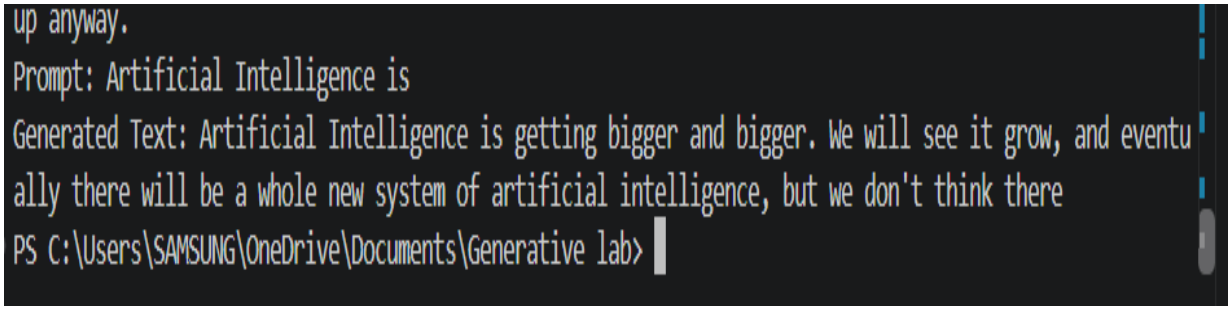
prompt = "Artificial Intelligence is"

result = generator(prompt, max_new_tokens=30, num_return_sequences=1)

print("Prompt:", prompt)

print("Generated Text:", result[0]["generated_text"])
```

Output



Experiment 20: Generate and Compare Responses for Multiple Prompts Using a Pre-trained LLM

Python Program

```
from transformers import pipeline

llm = pipeline("text-generation", model="gpt2")

prompts = [

    "Artificial Intelligence is",

    "Python programming is",

    "Climate change causes"

]

for prompt in prompts:

    result = llm(prompt, max_new_tokens=20,

                 do_sample=False, pad_token_id=50256)

    print("\nPrompt:", prompt)

    print("Response:", result[0]["generated_text"])Output
```

Output

```
PS C:\Users\SAMSUNG\OneDrive\Documents\Generative lab> & C:\Users\SAMSUNG\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/SAMSUNG/OneDrive/Documents/Generative lab/experiment20.py"
 years. It is a field that has
[transformers] Both `max_new_tokens` (=20) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

Prompt: Python programming is
Response: Python programming is a very complex language, and it's hard to understand how it works.

The main problem
[transformers] Both `max_new_tokens` (=20) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

Prompt: Climate change causes
Response: Climate change causes the climate to change, and the climate is changing.

The climate is changing because of human
PS C:\Users\SAMSUNG\OneDrive\Documents\Generative lab>
```